

BRUSSELS SOUTH CHARLEROI AP, Belgium

WMO: 064490

Lat: 50.4542N

Lon: 4.4400E

Elev: 187

StdP: 99.10

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-4.7	-9.9	1.7	-5.8	-7.5	2.0	-3.6	13.0	8.9	11.9	8.6	3.1	60	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)		(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	29.6	20.5	27.4	19.6	25.4	18.5	21.4	27.8	20.2	26.0	19.2	24.2	3.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.1	24.8	18.1	13.3	22.9	17.2	12.6	21.7	62.7	28.0	58.7	26.2	55.3	24.4	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.4	9.2	8.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-9.2	33.2	3.2	1.9	-11.5	34.5	-13.4	35.6	-15.2	36.6	-17.5	38.0		
			-9.7	23.3	3.2	1.4	-12.0	24.3	-13.8	25.1	-15.6	25.9	-17.9	26.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	3.0	3.8	6.5	9.8	13.3	16.3	18.5	18.2	14.9	11.3	6.9	3.8	(6)
(7)		DBStd	6.55	4.13	4.02	3.63	3.82	3.63	3.36	3.39	3.09	2.96	3.53	3.57	4.03	(7)
(8)		HDD10.0	892	218	173	116	49	9	0	0	1	27	104	194		(8)
(9)		HDD18.3	2948	475	406	366	257	160	81	41	41	110	218	344	450	(9)
(10)		CDD10.0	1099	1	1	9	43	113	188	263	253	148	68	10	2	(10)
(11)		CDD18.3	114	0	0	0	1	6	19	45	36	8	1	0	0	(11)
(12)		CDH23.3	958	0	0	0	9	49	170	374	305	49	2	0	0	(12)
(13)		CDH26.7	267	0	0	0	0	8	42	113	97	7	0	0	0	(13)
(14)	Wind	WSAvg	3.9	4.6	4.5	4.3	3.7	3.6	3.4	3.4	3.2	3.4	3.9	4.1	4.6	(14)
(15)	Precipitation	PrecAvg	878	82	71	62	53	66	76	80	88	60	70	76	94	(15)
(16)		PrecMax	1106	189	174	156	145	143	136	151	237	195	165	126	230	(16)
(17)		PrecMin	686	5	16	5	3	13	14	36	25	13	23	8	25	(17)
(18)		PrecStd	108	39	36	35	32	32	33	27	54	36	30	31	45	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	14.6	18.9	24.7	27.6	30.9	32.6	32.5	27.7	22.5	17.0	13.0	(19)
(20)		MCWB	10.6	9.0	11.6	15.8	19.2	21.0	20.7	21.4	19.4	16.7	13.3	11.3	(20)	
(21)		2%	DB	11.0	12.0	16.1	21.4	24.4	27.5	29.6	29.3	24.6	19.9	14.4	11.7	(21)
(22)		MCWB	9.6	8.7	10.6	13.6	17.1	19.7	20.3	20.7	18.2	15.7	12.0	10.3	(22)	
(23)		5%	DB	9.6	10.4	13.7	18.5	22.4	25.1	27.3	26.8	22.1	17.8	13.0	10.6	(23)
(24)		MCWB	8.3	8.4	9.8	12.4	15.8	18.4	19.4	19.4	17.1	14.7	11.3	9.4	(24)	
(25)		10%	DB	8.4	9.1	11.9	16.1	20.0	22.8	25.1	24.2	19.9	16.2	11.7	9.3	(25)
(26)		MCWB	7.3	7.6	9.0	11.3	14.7	17.2	18.5	18.2	15.7	13.8	10.2	8.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.6	12.7	16.7	20.2	22.1	22.5	23.3	20.5	17.4	14.0	11.7	(27)
(28)		MCDWB	11.9	12.5	17.1	23.3	26.3	28.9	29.5	30.4	25.6	20.8	16.1	12.6	(28)	
(29)		2%	WB	9.8	9.5	11.4	14.4	18.3	20.2	21.1	21.4	18.8	16.2	12.5	10.6	(29)
(30)		MCDWB	10.9	11.0	14.7	19.5	23.0	26.2	27.7	27.6	23.5	19.0	14.0	11.5	(30)	
(31)		5%	WB	8.4	8.7	10.3	12.9	16.6	19.0	20.0	20.0	17.6	15.1	11.5	9.6	(31)
(32)		MCDWB	9.4	10.1	13.0	17.3	20.6	23.8	26.0	25.4	21.1	17.2	12.9	10.6	(32)	
(33)		10%	WB	7.3	7.7	9.3	11.7	15.2	17.8	19.0	18.7	16.4	14.1	10.3	8.3	(33)
(34)		MCDWB	8.4	9.2	11.5	15.7	19.2	22.0	24.2	23.1	19.3	16.0	11.7	9.3	(34)	
(35)	Mean Daily Temperature Range		MDBR	4.3	5.2	6.8	8.4	8.6	9.0	8.9	7.8	6.5	4.7	4.1	(35)	
(36)		5% DB	MCDBR	4.9	6.7	10.1	12.0	12.3	12.7	12.6	13.0	10.9	8.2	5.9	4.5	(36)
(37)			MCWBR	4.4	4.8	6.1	6.3	6.3	6.4	5.4	5.9	5.6	4.8	4.3	3.9	(37)
(38)			MCDBR	4.8	5.6	8.4	10.6	10.8	11.3	11.3	11.8	9.5	7.5	5.2	4.3	(38)
(39)		5% WB	MCWBR	4.5	4.5	5.5	6.0	6.2	6.2	5.5	6.1	5.2	4.8	4.2	4.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.340	0.372	0.408	0.418	0.416	0.417	0.409	0.380	0.356	0.331	0.315	(40)	
(41)		taud	2.415	2.379	2.288	2.203	2.215	2.243	2.240	2.286	2.353	2.420	2.416	2.385	(41)	
(42)		Ebn at Noon	715	790	823	832	839	843	835	822	808	760	688	653	(42)	
(43)		Edh at Noon	62	83	107	130	135	132	131	119	100	79	62	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.79	1.46	2.62	4.14	4.91	5.36	5.10	4.34	3.29	1.94	0.93	0.62	(44)	
(45)		RadStd	0.11	0.29	0.37	0.48	0.50	0.49	0.60	0.47	0.34	0.16	0.11	0.10	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.73	N/A	N/A	N/A	N/A	(46)
(47)	Regional (78 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page